

CITYCARE GENERAL HOSPITAL

HOSPITAL FORMULARY STANDARD OPERATING PROCEDURE

Medicines Management, Formulary Governance & Drug Procurement

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TABLE OF CONTENTS

1	Purpose & Scope
2	Definitions & Abbreviations
3	Roles & Responsibilities
4	Formulary Structure & Drug Classification
5	Formulary Addition, Deletion & Review Process
6	Prescribing Standards & Restrictions
7	Drug Procurement, Storage & Stock Management
8	High-Alert & Controlled Drug Management
9	Non-Formulary Drug Requests & Exceptions
10	Audit, Pharmacovigilance & Best Practices

1. PURPOSE & SCOPE

This Standard Operating Procedure establishes the governance framework for the CityHospital Medicine Formulary — the authoritative, evidence-based list of approved medications for use within the hospital. It defines how drugs are classified, reviewed, procured, prescribed, dispensed, and monitored to ensure patient safety, therapeutic efficacy, and cost-effectiveness across all clinical departments.

- Define the structure, governance, and maintenance of the hospital formulary.
- Standardise prescribing practice across all clinical specialties and inpatient and outpatient settings.
- Ensure all medicines procured and stocked meet quality, safety, and regulatory standards.
- Establish clear processes for non-formulary drug requests and emergency access.
- Support pharmacovigilance, adverse drug reaction (ADR) monitoring, and continuous improvement.
- Comply with national medicines regulation, hospital licensing requirements, and accreditation standards.

NOTE: The Hospital Formulary is a living document. The most current version is held exclusively on the hospital intranet. Printed copies are uncontrolled and should not be used for prescribing decisions.

2. DEFINITIONS & ABBREVIATIONS

Term / Abbreviation	Definition
Formulary	The official hospital-approved list of medicines available for prescribing, procurement, and dispensing.
PTC	Pharmacy and Therapeutics Committee — the multidisciplinary governance body overseeing the formulary.
Non-Formulary Drug	A medicine not included in the current approved formulary; requires special authorisation.
High-Alert Medication	A drug that bears a heightened risk of causing significant patient harm if used in error.
Controlled Drug (CD)	A medicine subject to statutory control under national narcotics legislation (e.g., opioids, benzodiazepines).
ADR	Adverse Drug Reaction — any noxious, unintended response to a medicine at doses used in humans.
Generic Name	The internationally recognised non-proprietary name (INN) of a drug substance.
Brand Name	The proprietary name given to a drug by its manufacturer.
Off-Label Use	Use of an approved medicine outside its licensed indication, route, dose, or patient group.
Pharmacovigilance	The science and activities relating to the detection, assessment, and prevention of adverse effects of medicines.
LASA	Look-Alike Sound-Alike drugs — medicines whose names or packaging are similar and may cause confusion.
TDM	Therapeutic Drug Monitoring — measurement of drug concentrations to optimise dosing.
BNF	British National Formulary — reference guide for prescribing, dispensing, and administration of medicines.

3. ROLES & RESPONSIBILITIES

Role	Department	Key Responsibilities
PTC Chairperson (CMO or designate)	Executive / Clinical	Chair PTC meetings, final formulary approval, policy sign-off
Chief Pharmacist	Pharmacy	Formulary management, drug procurement, pharmacy team leadership
Clinical Pharmacologist	Pharmacy / Clinical	Drug evaluation, ADR analysis, TDM, clinical guideline input
Pharmacy & Therapeutics Committee	Multidisciplinary	Review additions/deletions, set restrictions, evaluate evidence
Ward Pharmacist	Pharmacy	Daily medicines reconciliation, formulary compliance, ADR reporting
Prescribing Physicians	All Specialties	Prescribe only formulary-approved drugs; seek authorisation for exceptions
Nursing Staff	All Wards	Safe drug administration, patient education, ADR identification
Procurement Officer	Supply Chain	Purchase only approved formulary drugs from licensed suppliers
Medical Records / IT	Information Systems	Maintain drug database, EMR drug library, alert system updates

4. FORMULARY STRUCTURE & DRUG CLASSIFICATION

The CityHospital Formulary is organised by therapeutic class in alignment with the WHO Model List of Essential Medicines and national formulary guidelines. All drugs are assigned a classification tier that determines availability, prescribing authority, and monitoring requirements.

4.1 Formulary Tiers

Tier	Classification	Availability	Prescribing Authority
Tier 1	Essential / Unrestricted	All wards and OPD; automatic stock	All registered medical officers
Tier 2	Restricted Use	Available on request; ward-level stock limited	Specialist physicians only (with designation)
Tier 3	Highly Restricted / Specialist	Dispensed per individual patient order only	Named consultant; PTC pre-approval may be required
Tier 4	Non-Formulary	Not stocked routinely	Requires non-formulary authorisation form and CMO sign-off

4.2 Therapeutic Categories in the Formulary

- **Section 1 — Cardiovascular:** Antihypertensives, antiarrhythmics, anticoagulants, antiplatelets, lipid-modifying agents.
- **Section 2 — Respiratory:** Bronchodilators, inhaled corticosteroids, mucolytics, oxygen therapy.
- **Section 3 — Infectious Diseases:** Antibiotics, antivirals, antifungals, antiparasitics (governed also by Antibiotic Policy SOP-PHARM-009).
- **Section 4 — Central Nervous System:** Analgesics, anaesthetics, antiepileptics, antipsychotics, anxiolytics.
- **Section 5 — Endocrine & Metabolic:** Insulin and oral hypoglycaemics, thyroid agents, steroids, calcium and electrolytes.
- **Section 6 — Gastrointestinal:** Antiemetics, antacids, proton pump inhibitors, laxatives, antidiarrhoeals.
- **Section 7 — Oncology & Immunology:** Chemotherapy agents, immunosuppressants, biological therapies (Tier 3 only).
- **Section 8 — IV Fluids & Nutrition:** Crystalloids, colloids, blood products, enteral and parenteral nutrition.
- **Section 9 — Dermatology, Ophthalmology & ENT:** Topical agents, eye drops, ear preparations.
- **Section 10 — Obstetrics & Gynaecology:** Uterotonics, tocolytics, hormonal agents, contraceptives.

4.3 LASA Drug Alert System

- All LASA drug pairs identified and flagged in the EMR prescribing system with a TALL MAN lettering alert.
- LASA drugs stored in separate, clearly labelled locations; never side-by-side on ward drug trolleys.
- LASA alert list reviewed and updated by PTC annually or after any LASA-related incident.

5. FORMULARY ADDITION, DELETION & REVIEW PROCESS

5.1 Submission of a New Drug for Formulary Consideration

- Step 1.** Any registered clinician, pharmacist, or department head may submit a New Drug Application (NDA) form to the Pharmacy Department.
- Step 2.** NDA form must include: generic and brand name, therapeutic indication, target patient population, evidence summary (minimum 2 RCTs or equivalent), proposed tier, cost analysis, and comparison with existing formulary alternatives.
- Step 3.** Chief Pharmacist performs initial administrative screening; incomplete applications returned within 5 working days.
- Step 4.** Complete applications tabled at the next scheduled PTC meeting (held quarterly or by emergency convening).
- Step 5.** PTC evaluates: clinical efficacy, safety profile, pharmacoeconomics, supply chain reliability, and formulary duplication.
- Step 6.** Decision communicated to applicant within 10 working days of PTC meeting: approved, rejected, or deferred pending further information.
- Step 7.** Approved drugs added to HIS drug library and EMR prescribing system within 30 days of approval.

5.2 Drug Deletion from Formulary

- Drugs may be removed for: safety concerns (regulatory withdrawal, safety signal), persistent non-availability, supersession by a safer or more cost-effective alternative, or very low utilisation.
- Deletion proposed to PTC; affected prescribers notified at least 30 days before removal from formulary.
- Patients currently on the drug to be deleted are reviewed; clinical team implements therapeutic substitution before deletion takes effect.

5.3 Annual Formulary Review

- Full formulary review conducted annually by the PTC using updated evidence and utilisation data.
- Each section reviewed by the relevant clinical specialty lead; recommendations submitted to PTC.
- Updated formulary published on hospital intranet; all staff notified via department heads and email bulletin.
- Previous version archived with version number and date; minimum 5-year archive period maintained.

6. PRESCRIBING STANDARDS & RESTRICTIONS

6.1 General Prescribing Standards

- All prescriptions must use the approved drug's generic (INN) name; brand prescribing permitted only for narrow therapeutic index drugs with documented clinical reason.
- Prescriptions must include: patient MRN, date, drug name, dose, route, frequency, duration, and prescriber signature with designation.
- Electronic prescribing via the EMR is mandatory for all inpatient and OPD settings; paper prescriptions accepted only during documented HIS downtime.
- Legibility: electronic prescriptions are preferred; all handwritten prescriptions must be legible; illegible prescriptions must be returned to prescriber before dispensing.
- Verbal or telephone orders accepted only in genuine clinical emergencies; must be documented in EMR within 1 hour by the prescriber.
- Weight-based dosing for paediatric patients: actual weight must be documented in EMR before prescription is issued.

6.2 Prescribing Restrictions by Tier

- **Tier 2 Restricted Drugs:** Prescribing limited to named specialties (e.g., vancomycin by infectious disease or microbiologist only). EMR will alert if a non-authorised prescriber attempts to order.
- **Tier 3 Highly Restricted Drugs:** Require an additional authorisation field in EMR; Chief Pharmacist or Clinical Pharmacologist counter-approval required before dispensing.
- **Chemotherapy agents:** Only oncology-trained prescribers may initiate; a double-check by second physician and pharmacist mandatory before dispensing.
- **Off-Label use:** Must be documented with evidence basis; patient informed consent obtained; documented in EMR; approved by Department Head.

6.3 Polypharmacy Review

- Patients on 8 or more regular medications flagged in EMR for mandatory clinical pharmacist review.
- Pharmacist uses the STOPP/START criteria for elderly patients to identify potentially inappropriate prescribing.
- Deprescribing recommendations communicated to the attending physician within 24 hours of flag.

7. DRUG PROCUREMENT, STORAGE & STOCK MANAGEMENT

7.1 Procurement Standards

- All drugs procured exclusively from suppliers holding a valid licence from the national medicines regulatory authority.
- Procurement Officer maintains an approved supplier list, reviewed and revalidated annually.
- Generic substitution permitted for Tier 1 drugs provided bioequivalence is demonstrated and documented.
- Emergency procurement from non-approved suppliers requires written authorisation from Chief Pharmacist and CMO; such drugs are quarantined pending verification.
- Cold-chain integrity documentation required for all temperature-sensitive medications (insulin, vaccines, biologicals).

7.2 Storage Standards

Storage Condition	Temperature Range	Examples	Monitoring Frequency
Ambient	15–25°C	Most oral tablets, capsules	Daily temperature log
Refrigerated	2–8°C	Insulin, vaccines, eye drops	Twice daily; alarms active
Frozen	-15 to -25°C	Selected biologicals	Daily; validated freezer
Controlled Drug Safe	15–25°C (locked)	Opioids, benzodiazepines	Per transaction + daily count

7.3 Stock Management

- Minimum and maximum stock levels set for each formulary drug based on 30-day consumption data.
- Automated reorder triggers activated in the pharmacy dispensing system when stock reaches minimum level.
- Expiry date checks conducted monthly; drugs within 3 months of expiry flagged for prioritised use or return to supplier.
- Expired or damaged drugs segregated in a quarantine area and disposed of per national pharmaceutical waste regulations.
- Drug wastage documented and reviewed monthly; excessive wastage triggers procurement pattern review.

8. HIGH-ALERT & CONTROLLED DRUG MANAGEMENT

8.1 High-Alert Medications

High-alert medications are those that bear a heightened risk of causing significant patient harm when used incorrectly. CityHospital designates the following drug categories as high-alert:

High-Alert Category	Examples	Special Safeguard Required
Anticoagulants	Warfarin, heparin, LMWH, DOACs	Double-nurse check; daily INR or anti-Xa monitoring
Insulin preparations	All insulins (basal, bolus, mixed)	Double-check by 2 nurses; blood glucose before administration
Concentrated electrolytes	KCl >2 mmol/ml, hypertonic saline	Never stored on ward; ICU only with physician supervision
Opioids / strong analgesics	Morphine, fentanyl, oxycodone	CD register; double-witness; count every shift
Neuromuscular blocking agents	Rocuronium, suxamethonium	Stored separately; red cap label; airway must be secured
Chemotherapy agents	All cytotoxic drugs	Oncology pharmacist only; double-check; PPE mandatory

8.2 Controlled Drug (CD) Management Workflow

- Step 1.** Physician prescribes CD in EMR and in the ward CD book with full details: drug, dose, route, frequency, quantity.
- Step 2.** Two registered nurses witness and verify CD from the CD safe; CD book signed by both nurses.
- Step 3.** CD administered to patient immediately; any portion wasted destroyed in the presence of the second witness and documented.
- Step 4.** CD safe key held by the nurse in charge; handover of keys documented at every shift change.
- Step 5.** CD count performed at every shift change; discrepancy triggers immediate incident report and pharmacy notification.
- Step 6.** CD register audited by pharmacy weekly; quarterly audit by the CMO-designated CD inspector.

WARNING: Concentrated potassium chloride (KCl >2 mmol/ml) must NEVER be stored on general wards. Only pre-diluted IV potassium solutions may leave the pharmacy. Any breach is a Level 4 Sentinel Event.

9. NON-FORMULARY DRUG REQUESTS & EXCEPTIONS

9.1 Non-Formulary Request Workflow

- Step 1.** Prescribing physician completes the Non-Formulary Drug Request (NFDR) form in the EMR, providing: patient MRN, clinical indication, evidence summary, formulary alternatives considered and reason for rejection, and expected duration of therapy.
- Step 2.** NFDR submitted to the ward clinical pharmacist for initial clinical review.
- Step 3.** Pharmacist approves or escalates to Chief Pharmacist within 4 hours (same-day decision for urgent cases).
- Step 4.** If approved: drug procured from approved external supplier; estimated delivery time communicated to prescriber.
- Step 5.** Drug dispensed as a single-patient supply; not added to ward stock.
- Step 6.** Non-formulary drug use documented in EMR; utilisation reviewed by PTC at next quarterly meeting.

9.2 Emergency Non-Formulary Access

- Life-threatening clinical emergencies: Chief Pharmacist or on-call pharmacist may authorise verbal approval; NFDR completed retrospectively within 24 hours.
- Out-of-hours requests: on-call pharmacist holds authority to approve Tier 2 non-formulary drugs; Tier 3 requires on-call consultant and on-call pharmacist joint decision.
- All emergency non-formulary events reported to PTC monthly for trend analysis.

9.3 Off-Label Drug Use

- Off-label use requires documented evidence basis and is classified as non-formulary unless the PTC has pre-approved the specific off-label indication.
- Informed consent discussing off-label status must be obtained from the patient or guardian; documented in EMR.
- Off-label prescribing by trainees requires consultant co-signature.
- All off-label ADRs must be reported to the national medicines regulatory agency pharmacovigilance programme.

10. AUDIT, PHARMACOVIGILANCE & BEST PRACTICES

10.1 Formulary Compliance Audit

Audit Parameter	Target	Frequency	Responsible
Prescriptions using generic names	Greater than 95%	Monthly	Ward Pharmacist
Non-formulary requests with complete NFDR	100%	Monthly	Chief Pharmacist
CD count discrepancies	Zero tolerance	Per shift	Charge Nurse + Pharmacy
High-alert double-check compliance	100%	Weekly spot audit	Clinical Pharmacist
Expiry date compliance	Zero expired drugs in active stock	Monthly	Pharmacy Technician
LASA drug storage separation	100%	Monthly	IPC + Pharmacy

10.2 Adverse Drug Reaction (ADR) Reporting

- All ADRs identified by any clinical staff must be reported via the EMR ADR reporting module within 24 hours of identification.
- Serious ADRs (life-threatening, requiring hospitalisation, or causing permanent disability) reported to the national regulatory authority within 15 days.
- Fatal ADRs or unexpected serious ADRs: immediate notification to Chief Pharmacist and CMO; report to regulator within 7 days.
- ADR report triggers clinical pharmacologist review; causality assessed using the Naranjo Algorithm.
- Monthly ADR summary circulated to all prescribers; quarterly review by PTC.

10.3 Medication Error Reporting & Learning

- All medication errors (prescribing, dispensing, administration, monitoring) reported as clinical incidents in the EMR.
- Near-miss medication errors reviewed at monthly Medication Safety Committee meeting.
- RCA performed for all medication errors causing patient harm; CAPA implemented within 30 days.
- Medication safety learning bulletins distributed to all clinical staff after significant events.

10.4 Best Practices

- 5 Rights of medication administration verified before every drug administration: Right Patient, Right Drug, Right Dose, Right Route, Right Time.
- Barcode medication administration (BCMA) system used at the bedside to eliminate transcription and administration errors.
- Patient education on discharge medications includes: drug purpose, dose, timing, side effects, and monitoring requirements.
- Regular clinical pharmacy rounds: pharmacist attends ward rounds 3 days per week; reviews all prescriptions for safety, interactions, and appropriateness.
- Annual 'Medicine Safety Week': hospital-wide campaign promoting safe prescribing, LASA awareness, and ADR reporting culture.